

CARL LOUIE A. QUILLO

SOFTWARE ENGINEER | WEB DEVELOPER

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PROFESSIONAL SUMMARY

Information Technology graduate with hands-on experience delivering production features using React, TypeScript, Node.js, and MongoDB within Agile/Kanban teams. During a 500-hour internship, merged 15+ development tasks into a live employee management system, covering UI development, REST API integration, debugging, and version control. Independently engineered an IoT-based real-time monitoring system (ESP32 + React + Node.js) as a capstone project. Strong foundation in data structures, databases, networking, and the full SDLC. Seeking an Engineer, Full-Stack Developer, or Web Developer role to contribute immediately and grow within a collaborative engineering team.

EDUCATION

Iloilo Science and Technology University (2022-2026)

Bachelor of Science in Information Technology(BSIT).

Iloilo National High School(2015-2021)

Technical-Vocational-Livelihood - Information and Communication Technology(TVL-ICT)

SKILLS

- **Programming Languages:**
 - JavaScript, TypeScript, Python, Java, C/C++, PHP, HTML5, CSS3
- **Frameworks & Libraries:**
 - React, Node.js, Tailwind CSS, Material UI (MUI)
- **Databases:**
 - MongoDB, MySQL, PostgreSQL
- **Tools & Platforms:**
 - Git, GitLab, GitHub, VS Code, MS Office, Cisco Packet Tracer, VirtualBox
- **Concepts & Practices:**
 - REST APIs, Agile/Kanban, SDLC, OOP, Responsive Design, Unit Testing & Debugging, Version Control
- **Networking & Hardware:**
 - VLAN Routing, LAN/Wi-Fi Configuration, PC Assembly, ESP32/Arduino Microcontrollers

LANGUAGE SKILLS

- English
- Filipino
- Hiligaynon

WORK EXPERIENCE

Kingsland Technology Philippines (Feb 2026 – May 2026)

IT Intern / Web Development Intern

- Built and maintained features for an internal employee management system used organization-wide, working across the full stack with React, TypeScript, Node.js, and MongoDB.
- Designed and implemented reusable UI components with React, Tailwind CSS, and Material UI (MUI), improving usability and front-end consistency.
- Engineered a multi-country holiday calendar feature, extending system functionality for international teams.
- Diagnosed and resolved recurring application bugs across the platform, improving system reliability and user experience.
- Integrated and consumed REST APIs to support core application features and data management requirements.
- Collaborated in Agile/Kanban sprints, participating in sprint planning and daily stand-ups to deliver tasks on schedule.
- Used GitLab for version control, branching, and merge requests in a multi-developer, collaborative codebase.
- Completed and merged 15+ development tasks into production, following full SDLC practices planning, development, testing, documentation, and deployment.

La Paz Public Market(2012 – 2016)

Retail Clerk — Family-Owned Store

- Handled 50+ daily sales transactions in a fast-paced public market environment while ensuring accuracy in cash handling and record-keeping.
- Maintained and updated digital and manual inventory systems, improving stock tracking accuracy and reducing discrepancies.
- Assisted in product display and organization to improve customer accessibility and sales visibility.
- Resolved customer inquiries and handled complaints professionally, ensuring customer satisfaction and repeat business.
- Coordinated with suppliers for restocking and delivery schedules, helping maintain consistent inventory levels.
- Supported daily store operations including opening/closing procedures, cash reconciliation, and basic reporting.

ACADEMIC PROJECTS

SolarCharge: Solar Monitoring & Remote Power Management
System Administration Simulation
Student Grade Analysis System
Ascend - Project mockup

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ACADEMIC PROJECTS

SolarCharge: Solar Monitoring & Remote Power Management System

Capstone Group Project (5)

- Designed and developed SolarCharge, an IoT-based smart solar charging and power management system integrating ESP32 devices with a full-stack web application.
- Developed the web application using **React, Node.js, Express.js, TypeScript, and Supabase**, providing responsive dashboards for monitoring charging stations and managing connected devices.
- Designed and implemented the database architecture using Supabase/PostgreSQL, including user accounts, charging stations, ports, subscriptions, and system-related data.
- Developed REST APIs with Node.js and Express.js to handle communication between the web application, backend services, and IoT devices.
- Implemented user authentication and access control using Supabase Auth, with separate user and administrative functionality.
- Integrated MQTT/EMQX messaging for communication between the backend and ESP32 charging stations, enabling remote monitoring and device control.
- Implemented real-time monitoring and data synchronization for charging and power-related metrics, with responsive interfaces accessible from desktop and mobile devices.
- Integrated PayPal subscription payments for premium charging access and designed the application flow for free and premium charging ports.
- Developed ESP32 firmware and integrated **ACS712 current sensors, relay modules, and charging hardware** for power monitoring and remote switching.
- Created system architecture diagrams, database/ERD designs, sequence diagrams, API flows, and technical documentation following software engineering practices.

System Administration Simulation

Academic Group Project (2)

- Planned and implemented a simulated University network environment for system administration and IT infrastructure management.
- Configured Windows, including over 200+ user account management, file permissions, network services, and system security settings.
- Performed software installation, system maintenance, troubleshooting, backup procedures, and basic server administration tasks.
- Applied networking concepts such as IP addressing, DNS, and DHCP while documenting configuration procedures and operational workflows.

Student Grade Analysis System

Academic Group Project (5)

- Designed and developed a Python-based application for analyzing and visualizing student academic performance.
- Processed and cleaned grade datasets by handling missing values, removing duplicates, and standardizing data.
- Generated descriptive statistics and correlation analyses to identify relationships between subjects and student performance.
- Created interactive visualizations including histograms, box plots, heatmaps, bar charts, and scatter plots to present academic trends.
- Supported importing, merging, and exporting datasets to simplify educational data analysis workflows.

Ascend - Project mockup

Academic Group Project (5)

- Designed and documented a conceptual system mockup for “Ascend,” applying structured Systems Analysis and Design (SAD) principles.
- Conducted requirement analysis, identified user roles, and defined system objectives and constraints.
- Created process models including use case diagrams, data flow diagrams (DFD), and basic system architecture to represent system behavior.
- Developed UI/UX wireframes and workflow mockups to visualize user interaction and system navigation.
- Produced a complete project documentation set covering system requirements, proposed solutions, and design justification.

Basic Programming

Academic Projects

- Java Application – “Tipon” (Personal Finance Management System)
 - Designed and developed a basic personal finance management application using Java, focusing on budgeting, expense tracking, and simple data handling.
- C Application – BMI Calculator
 - Created a simple command-line program in C that calculates Body Mass Index (BMI) based on user input and outputs health category results.
- PHP – Personal Portfolio Website
 - Built a basic dynamic portfolio website using PHP to display personal information, projects, and contact details.
- HTML/CSS – Static Website Project
 - Developed a responsive static website using HTML and CSS, focusing on layout design, structure, and user interface fundamentals.